



NATIONAL RESEARCH CENTRE

«KURCHATOV INSTITUTE»

«КУРЧАТОВ ИНСТИТУТ»

Kurchatov synchrotron radiation source from the 2nd to the 4th generation



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October 01-05, 2018

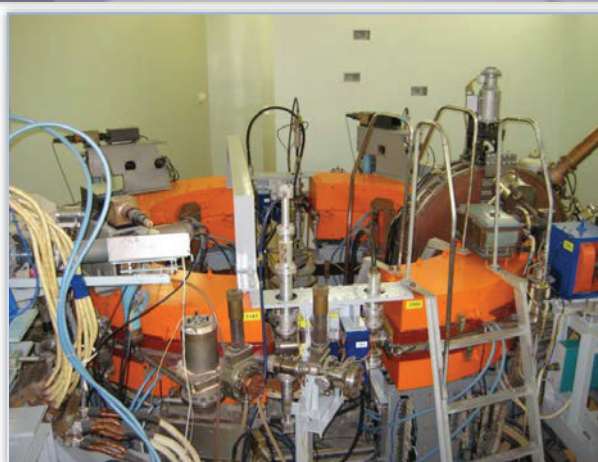
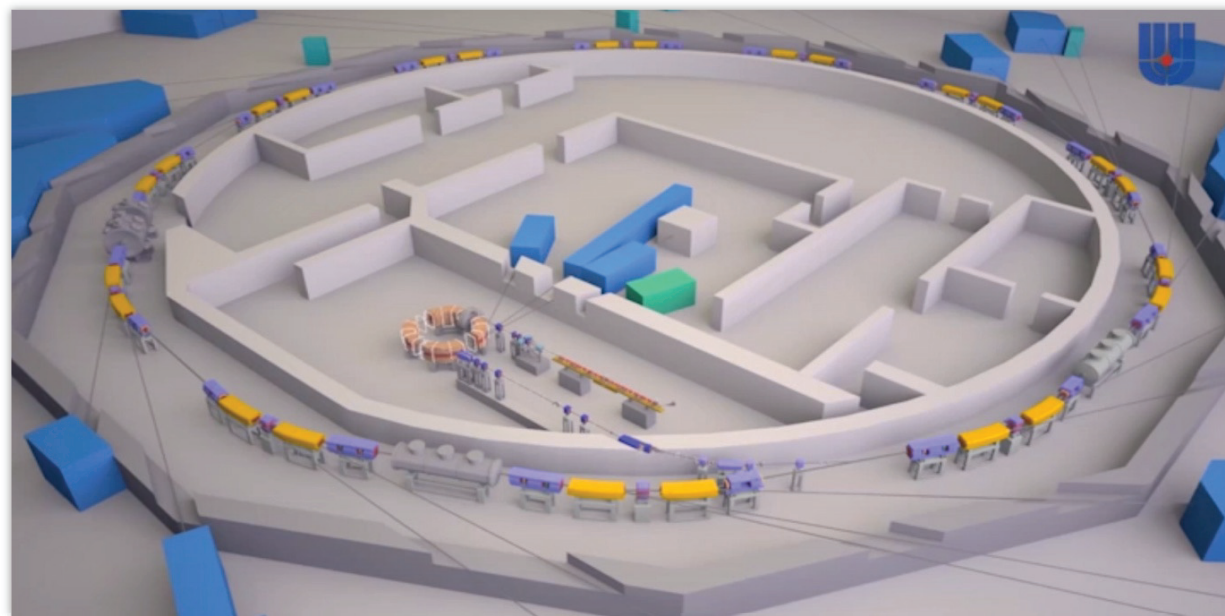


Outline

- Kurchatov synchrotron radiation source current status
- Possibilities for the 3rd generation upgrade
- USSR Project – 4th generation source
- Conclusions

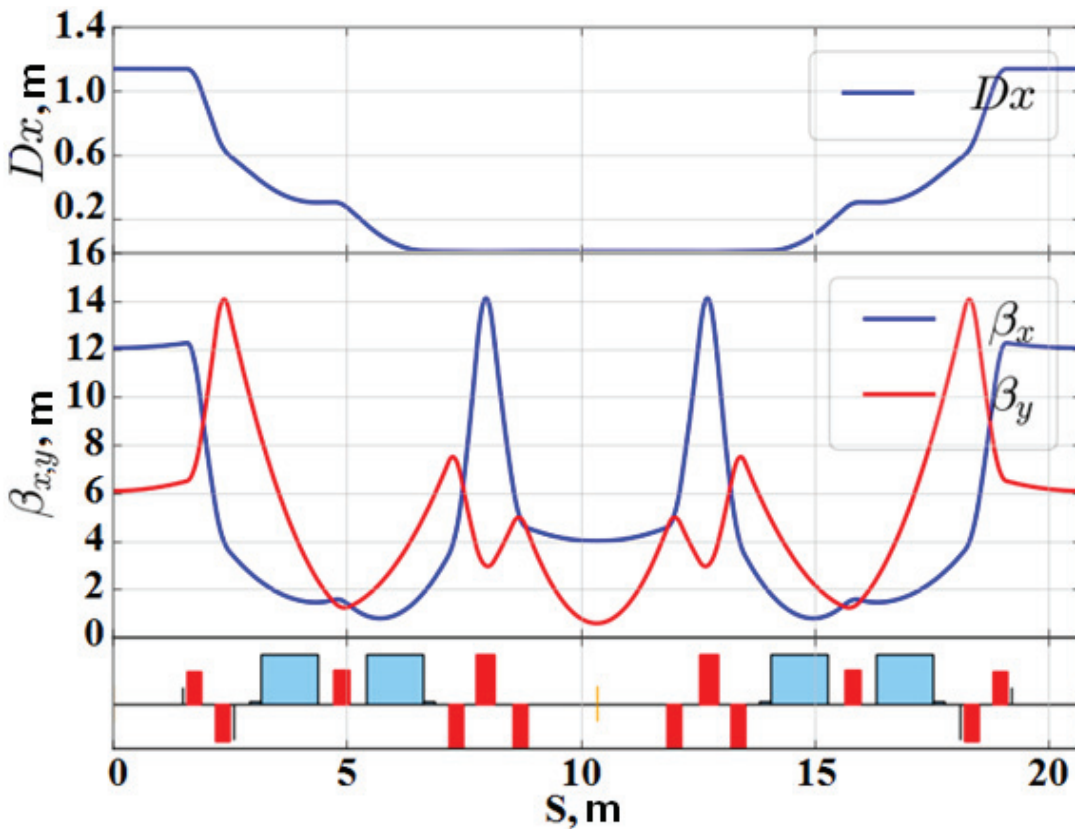


Kurchatov synchrotron radiation source



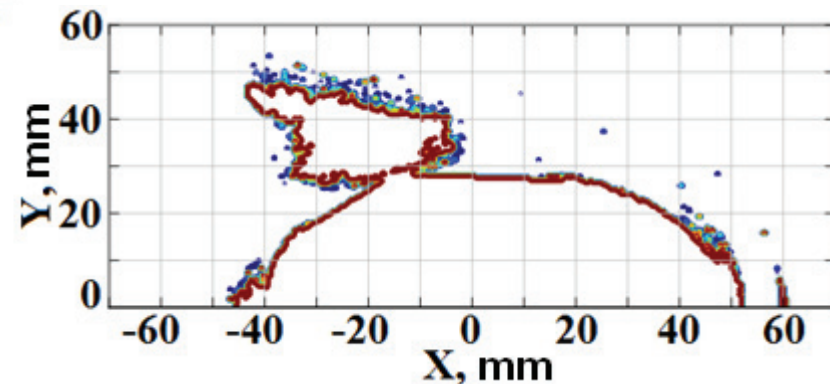


Kurchatov synchrotron radiation source



The main parameters

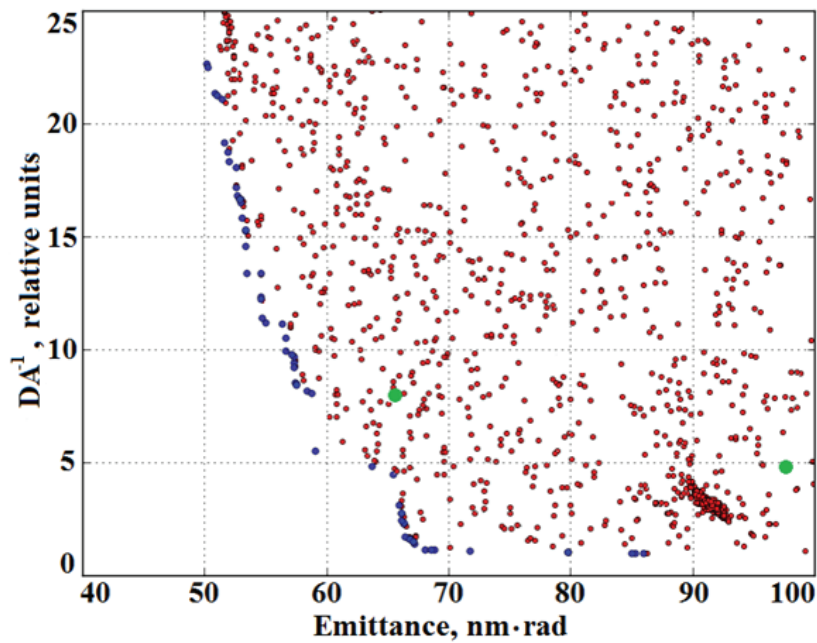
Energy	2.5 GeV
Circumference	124.13 m
Superperiods	6
Beam current	100 – 150 mA
Beam lifetime	15-20 h
Emittance	98 nm·rad
Tunes	7.775 / 6.695
Chromaticity	-16.9 / -12.9



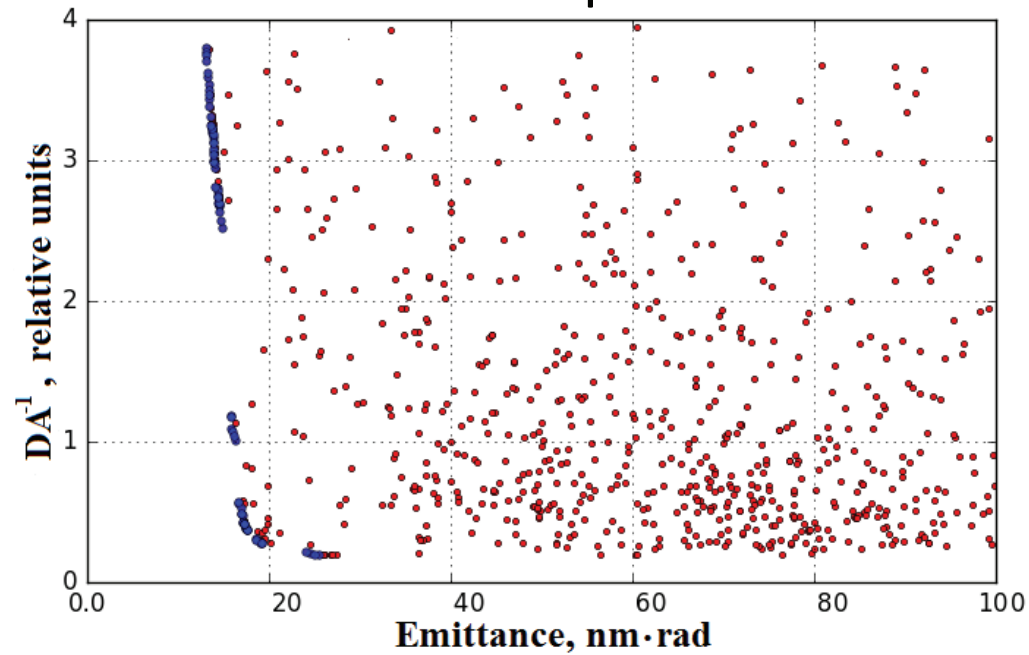


Possibilities for upgrade

Zero dispersion



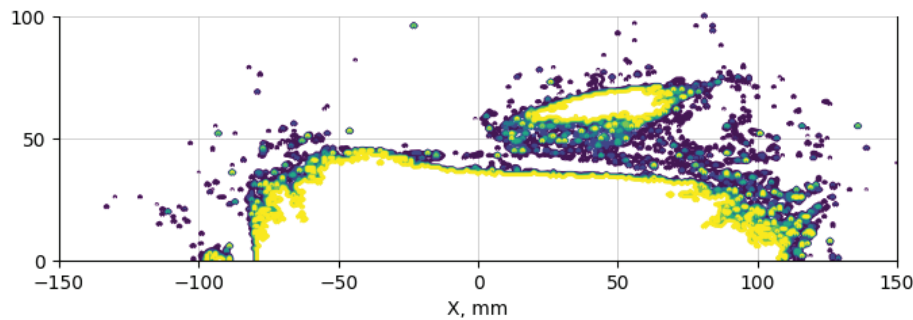
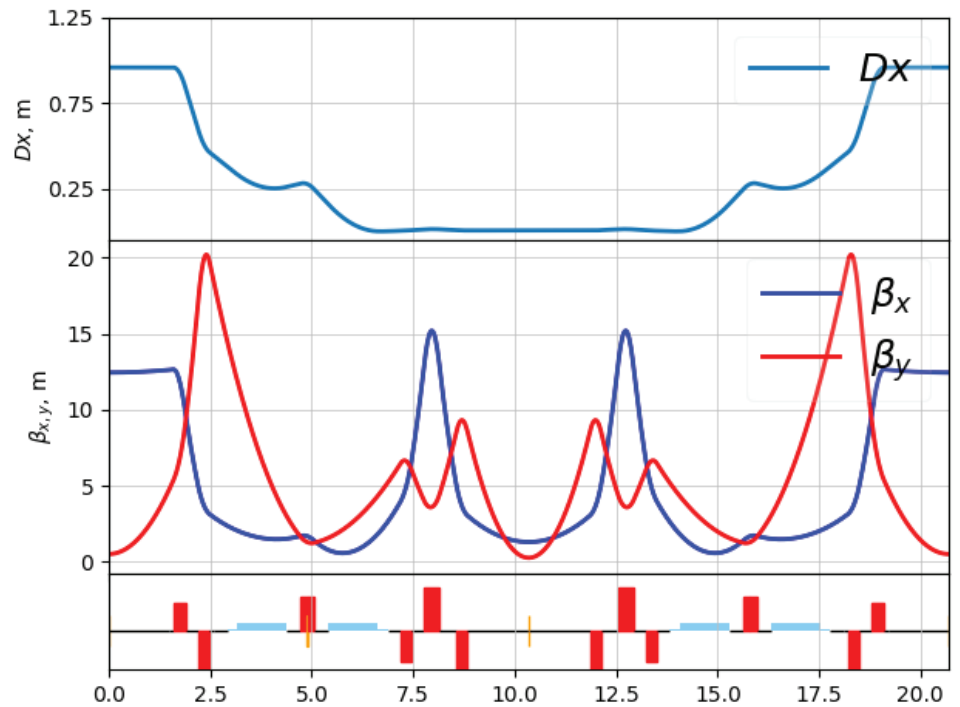
Non zero dispersion



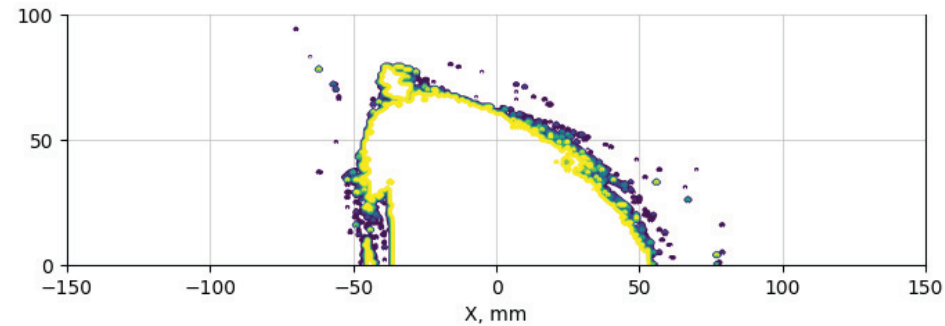
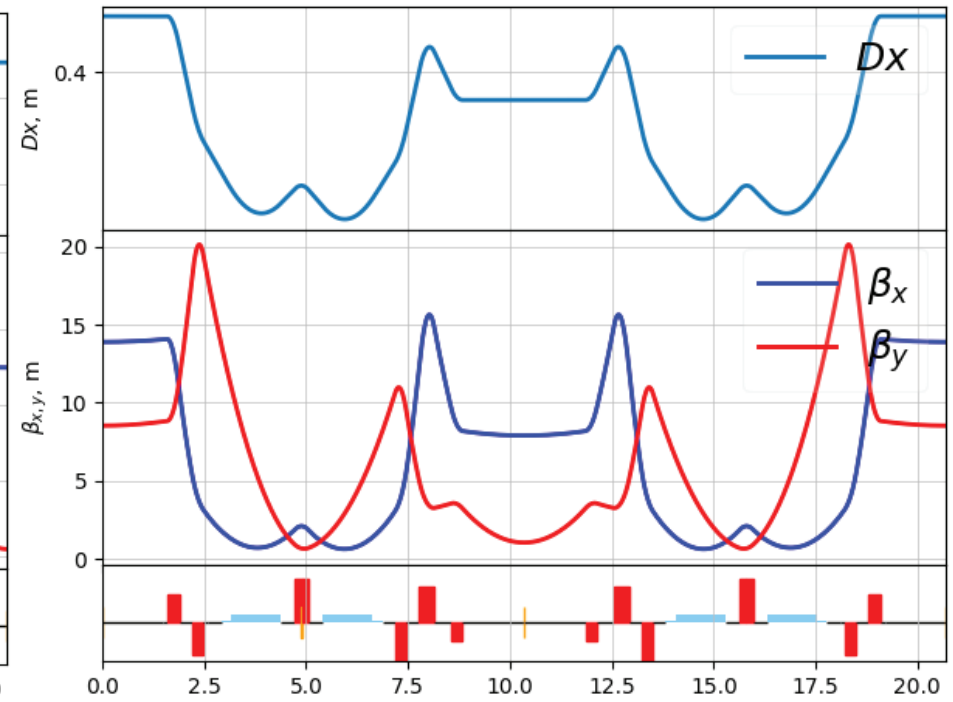


New low emittance lattices

67 nm·rad lattice



17 nm·rad lattice

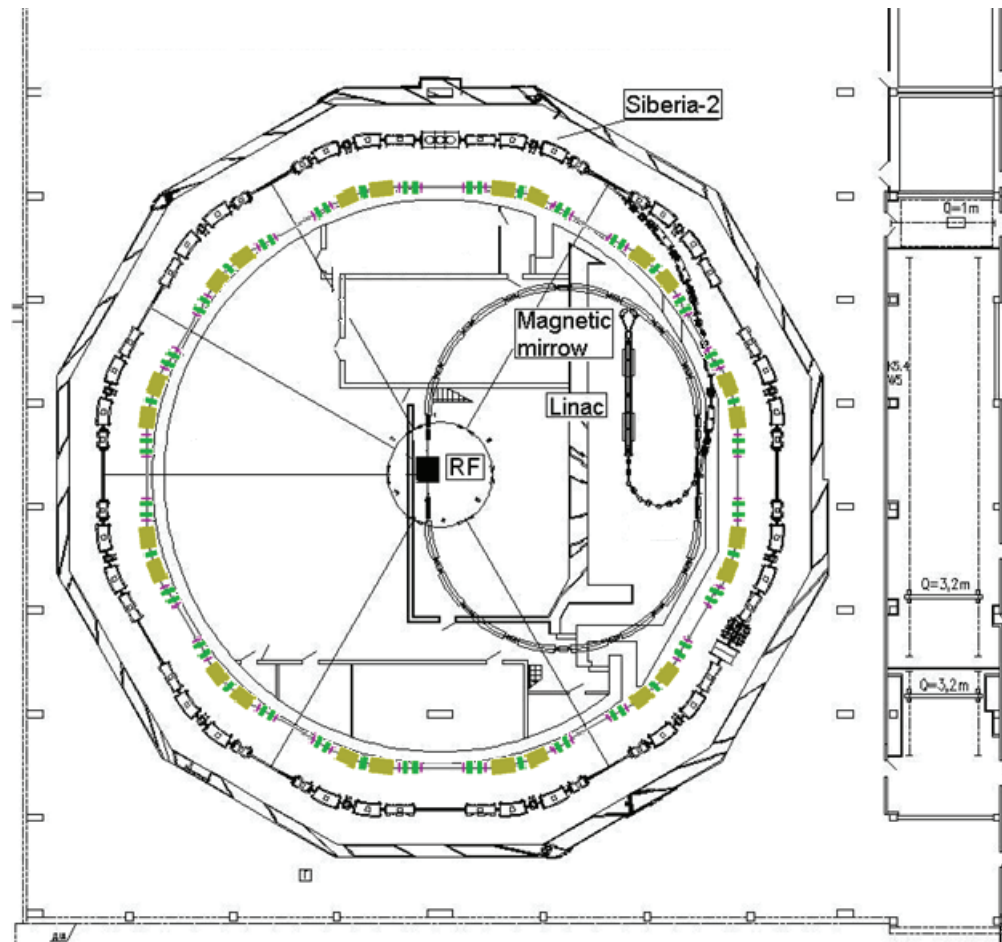




Full energy booster synchrotron

Energy: 80/160 - 2500 MeV

Emittance: 50-90 nm·rad





The most progressive sources



MAX-IV 3 GeV 200 pm 528 m



ESRF EBS 6 GeV 140 pm 844 m



SLS-2 2.4 GeV 125 pm 290 m

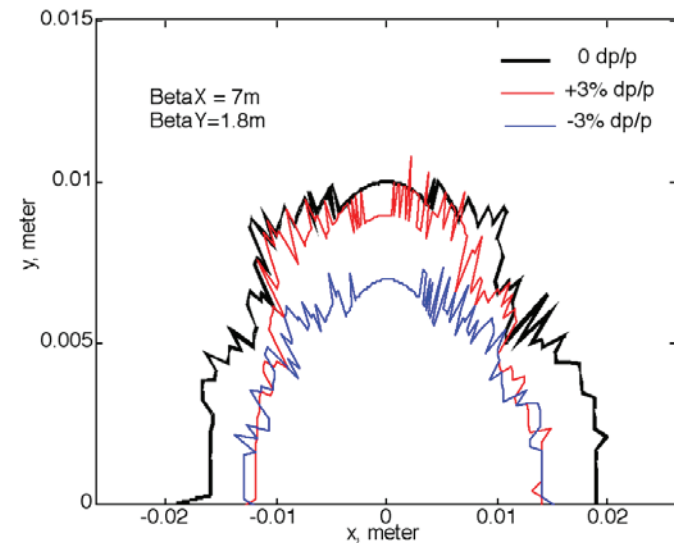
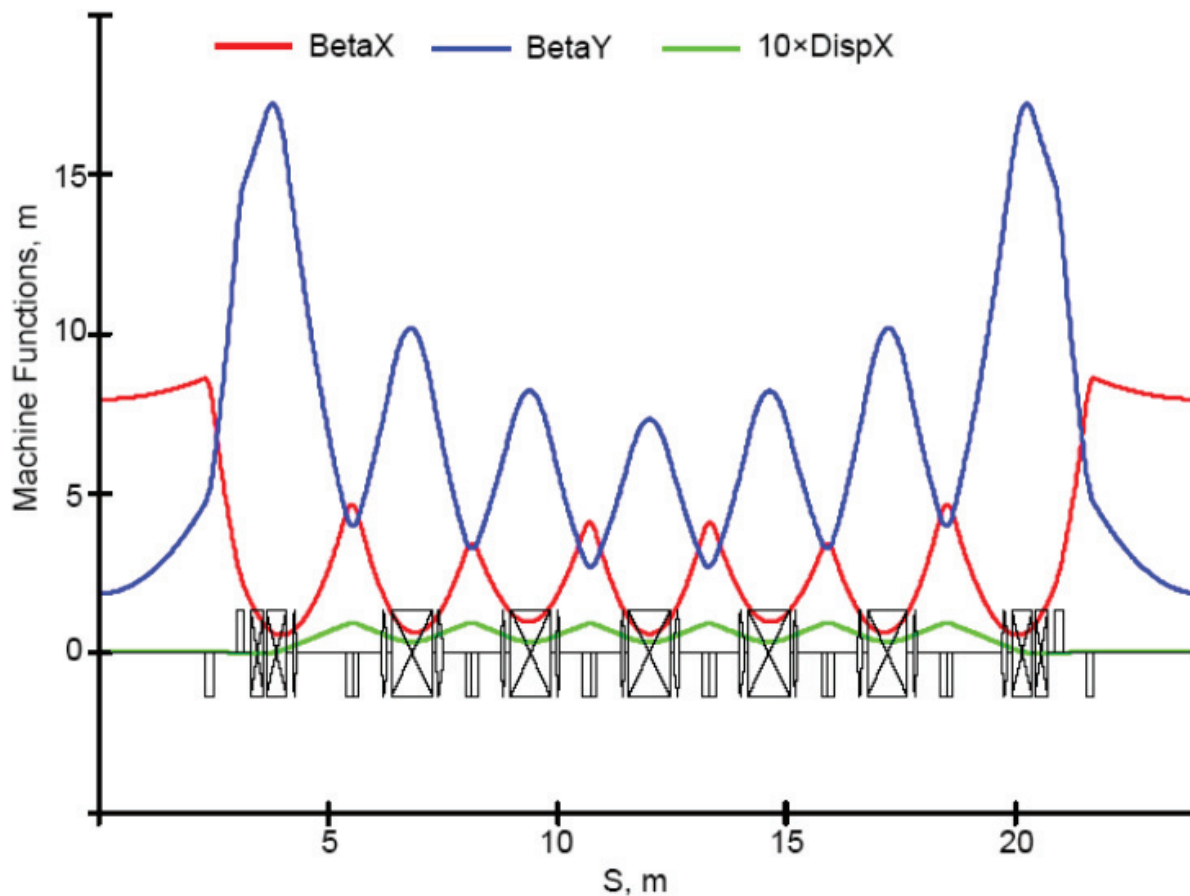


MAX-IV

Energy: 3 GeV

Circumference: 528 m

Emittance: 200 pm·rad



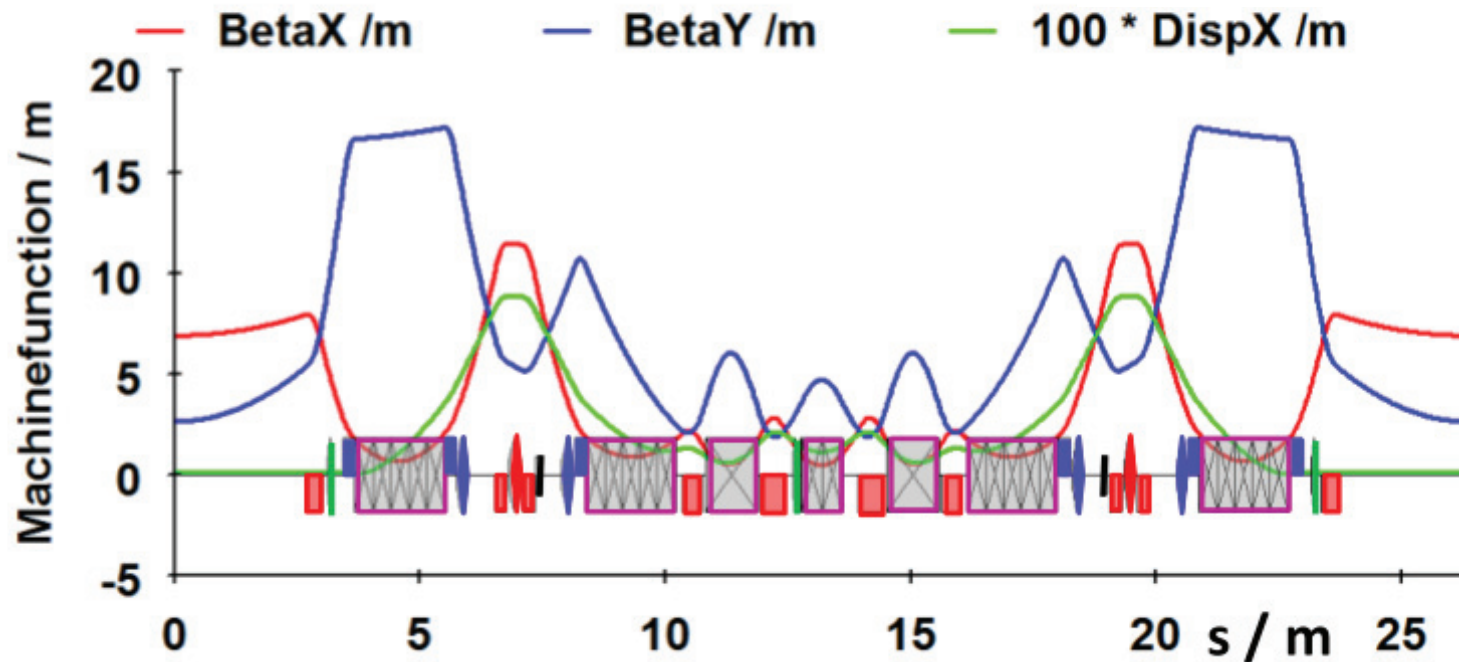
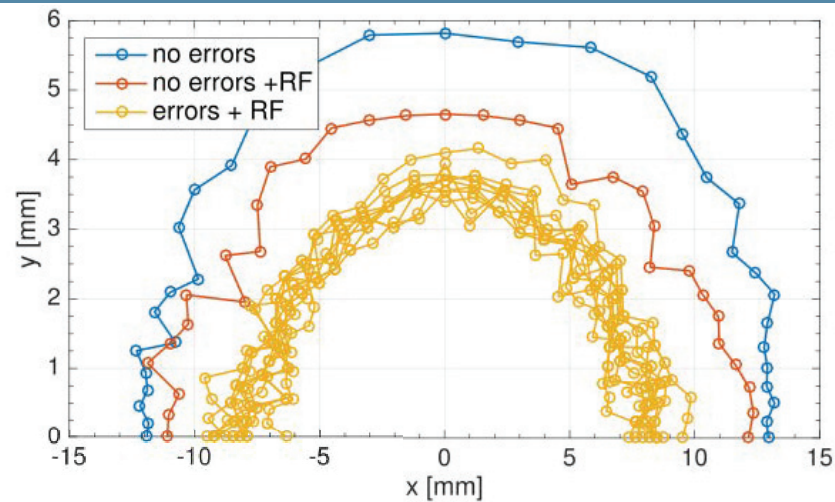


ESRF EBS

Energy: 6 GeV

Circumference: 844 m

Emittance: 140 pm·rad



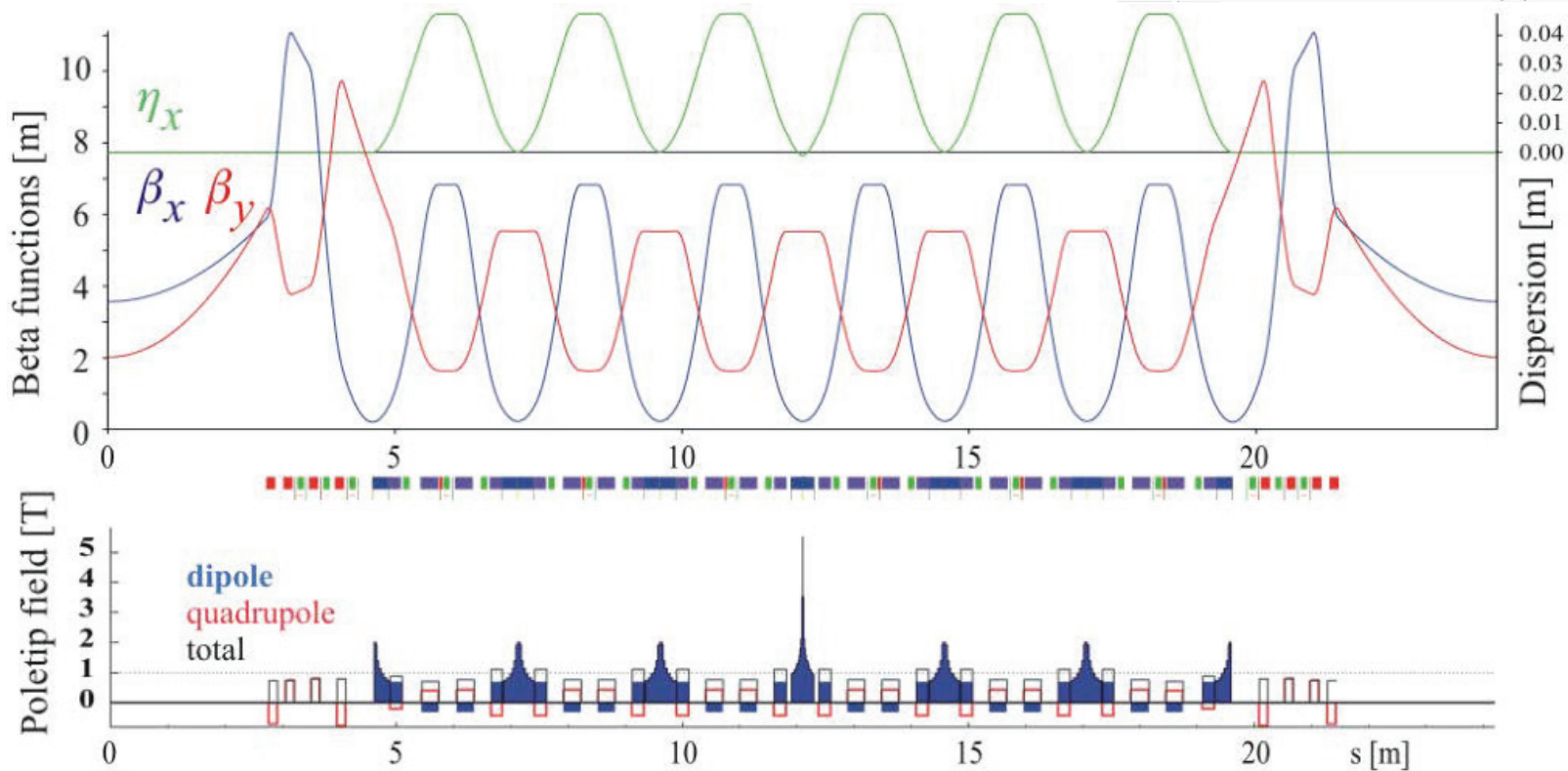
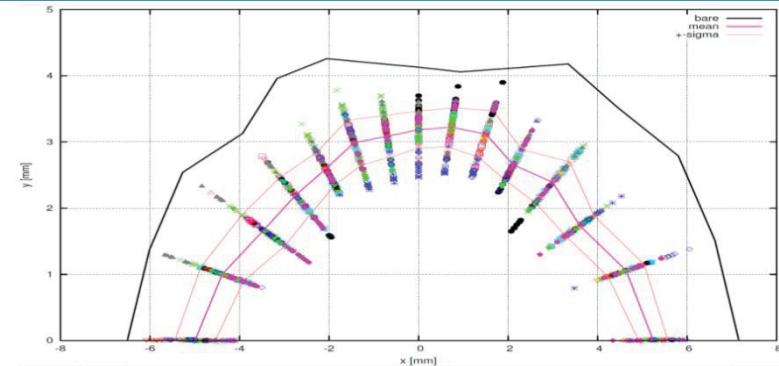


SLS-2

Energy: 2.4 GeV

Circumference: 290 m

Emittance: 125 pm·rad



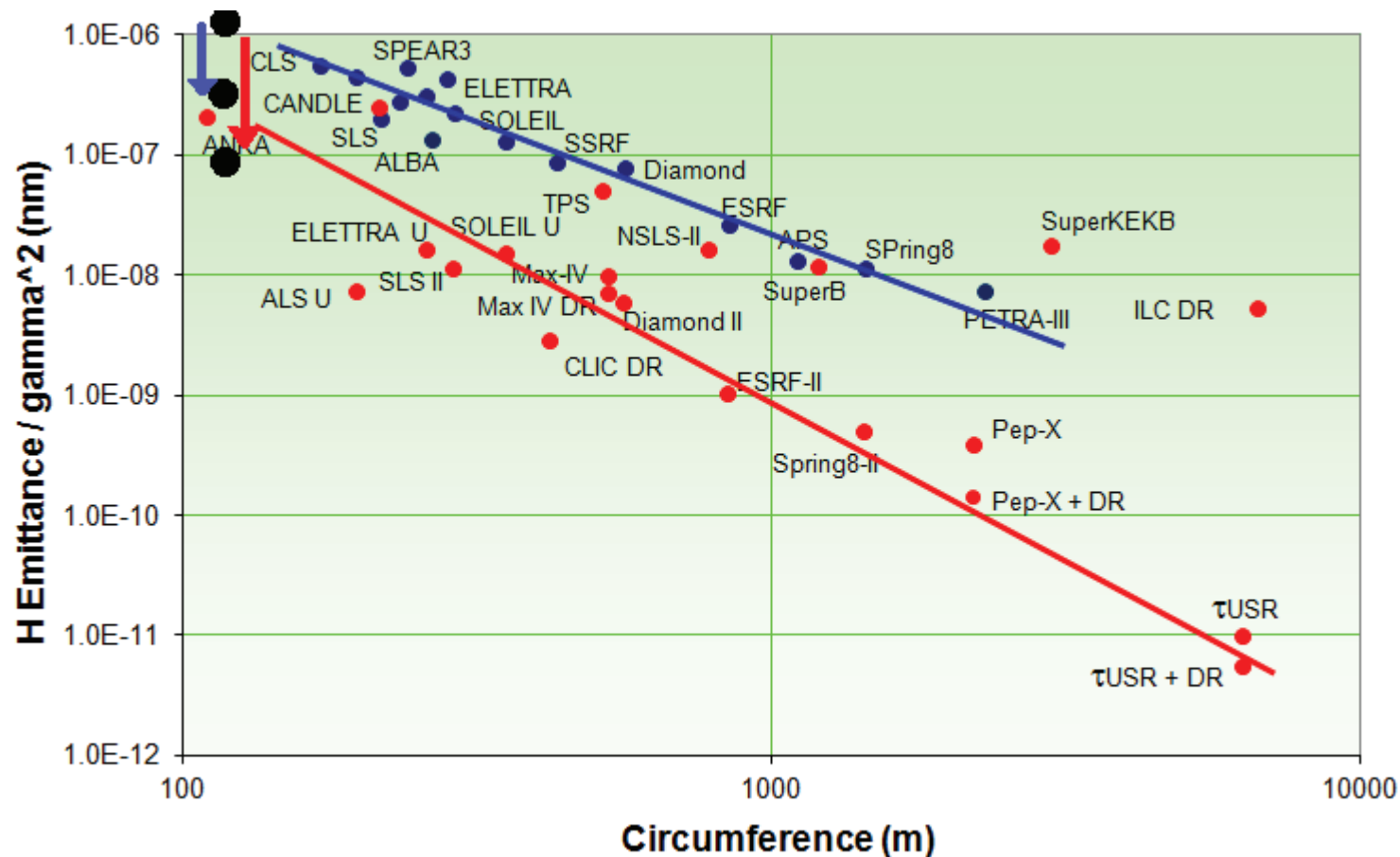


Towards the 3rd generation source

Lattice: MBA

Emittance: 1-5 nm·rad (0.5 nm·rad?)

Energy: 2.5 GeV (2.0-2.5 GeV?)





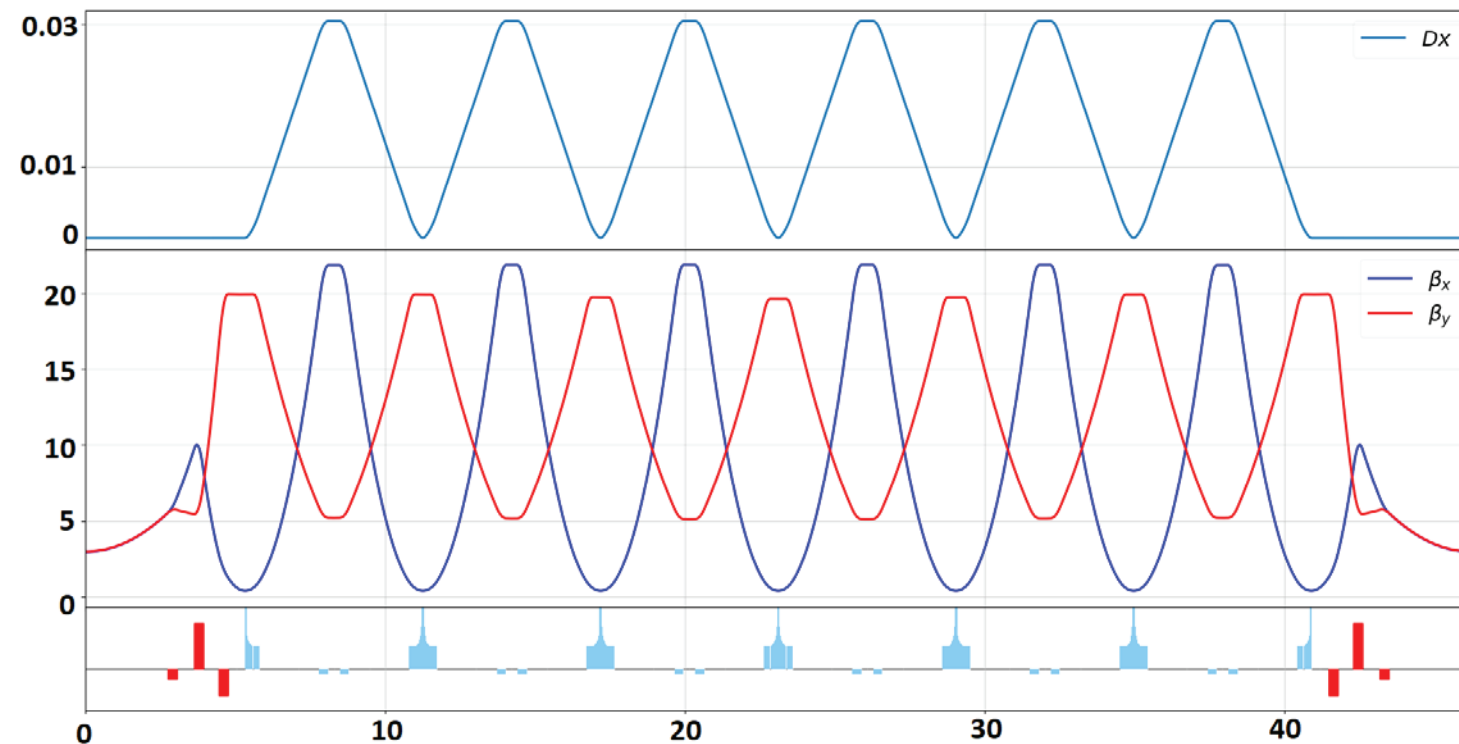
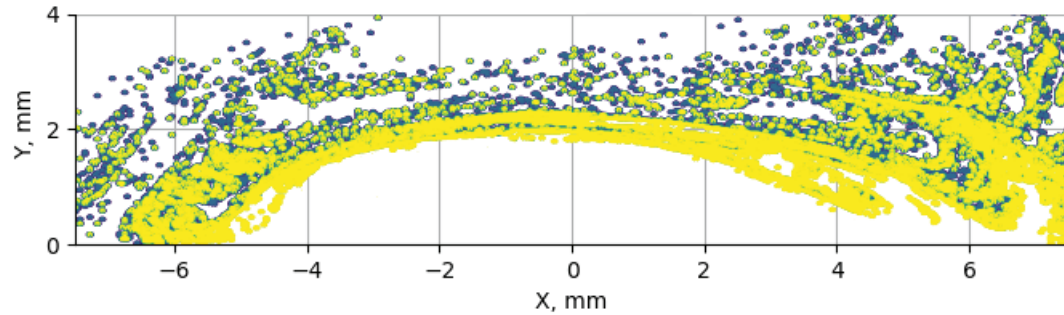
The 4th generation source basic requirements

- Diffraction-limited source at a wavelength of 1 Å
- Full spatial coherence
- Maximum achievable temporal coherence without monochromatization
- Average source brightness $> 10^{23-24}$ photons $\cdot$ s⁻¹ $\cdot$ mm⁻² $\cdot$ mrاد⁻² $\cdot$ ($\Delta\lambda/\lambda=0.1\%$)⁻¹
- Peak source brightness $> 10^{33}$ photons $\cdot$ s⁻¹ $\cdot$ mm⁻² $\cdot$ mrاد⁻² $\cdot$ ($\Delta\lambda/\lambda=0.1\%$)⁻¹
- High long-term stability
- Simultaneous work of a large number of user stations



USSR Project

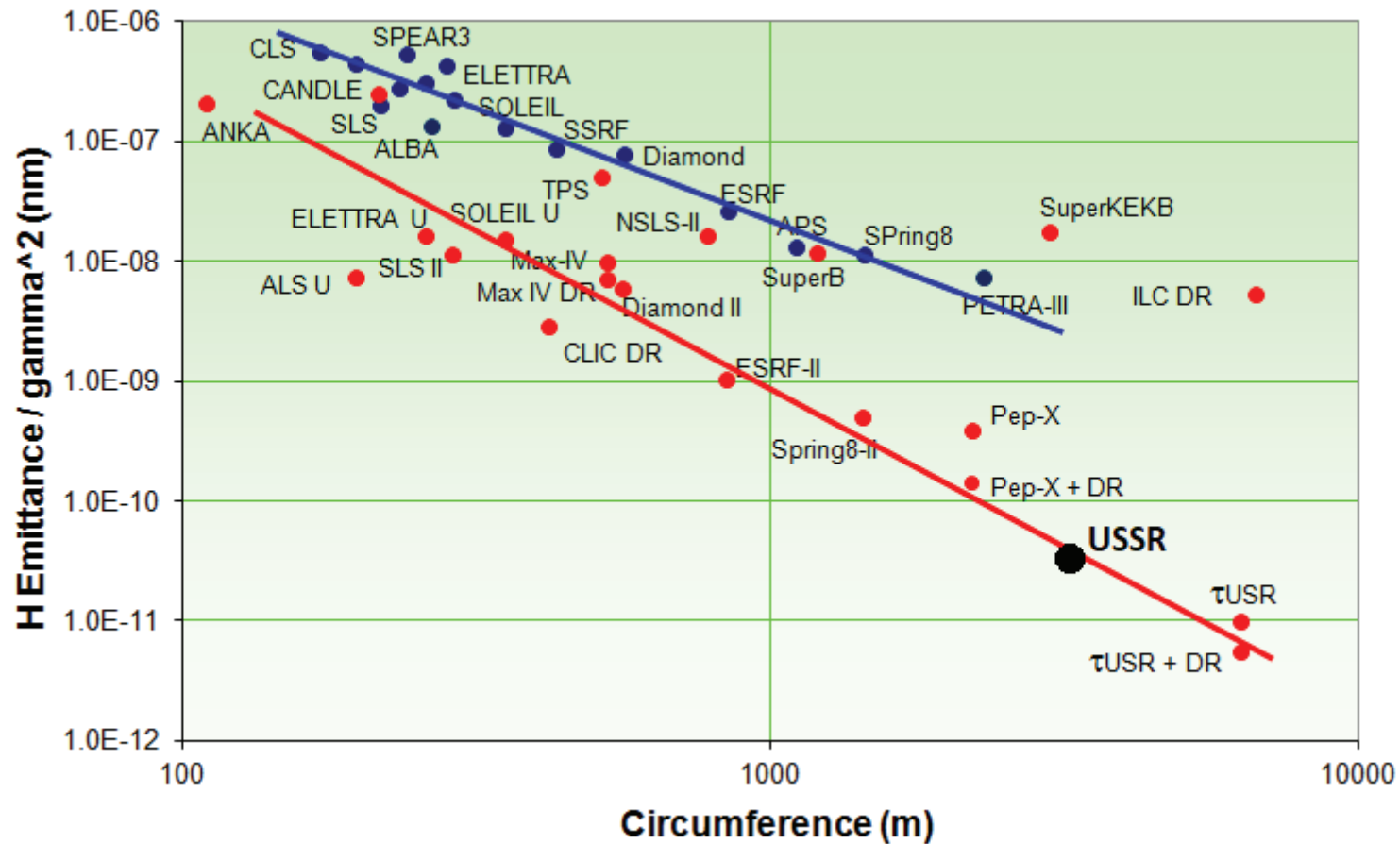
Energy: 6 GeV
Circumference: 2218 m
Superperiods: 48
Emittance: 10.42 pm·rad



at injection point
 β_x : 24.3 m
 β_y : 4.25 m

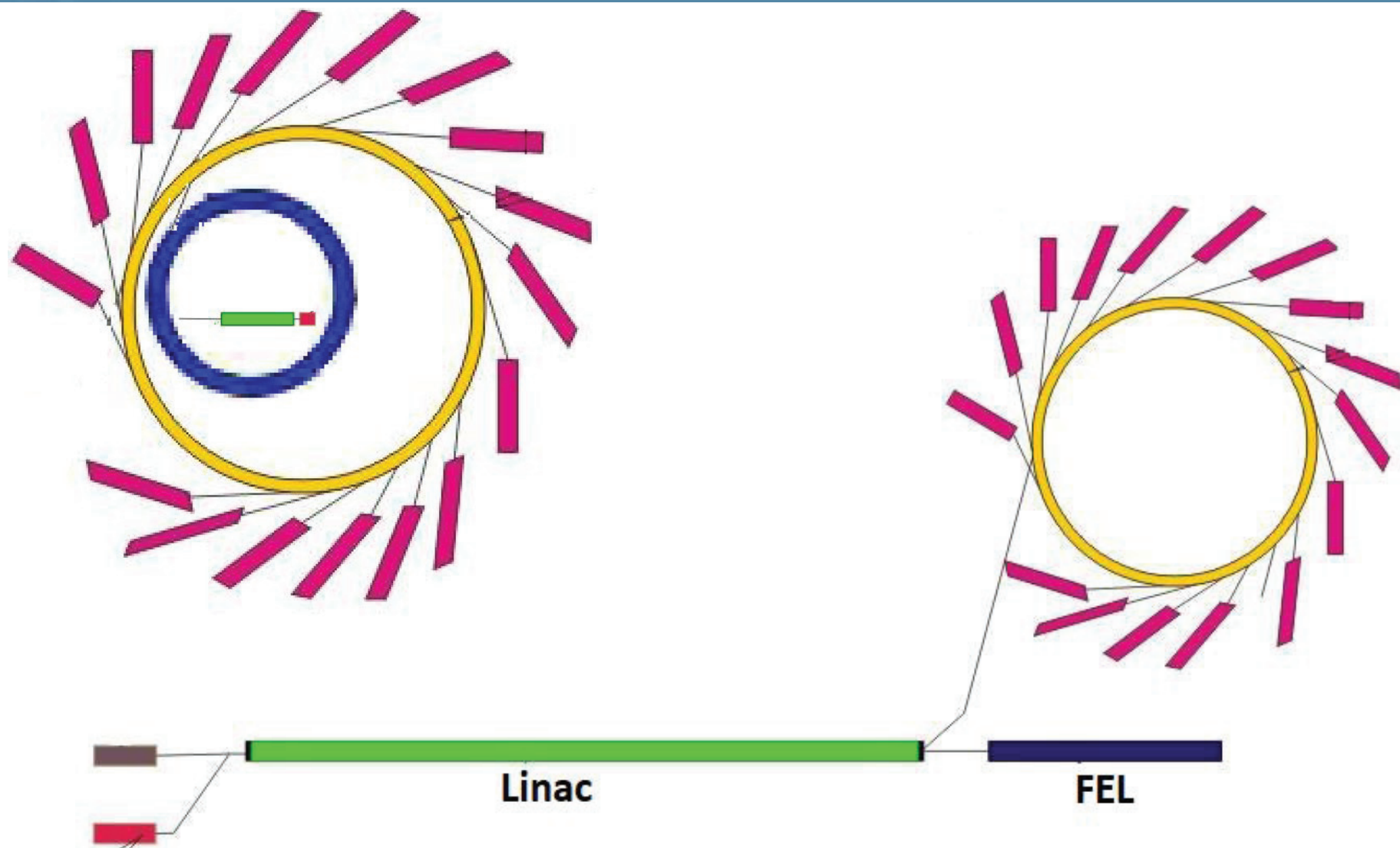


USSR Project





USSR injection



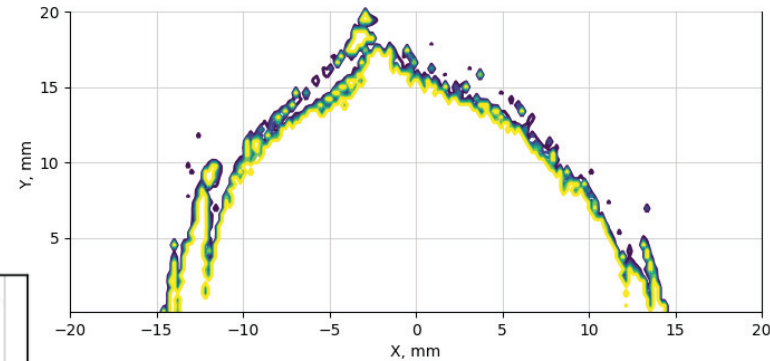
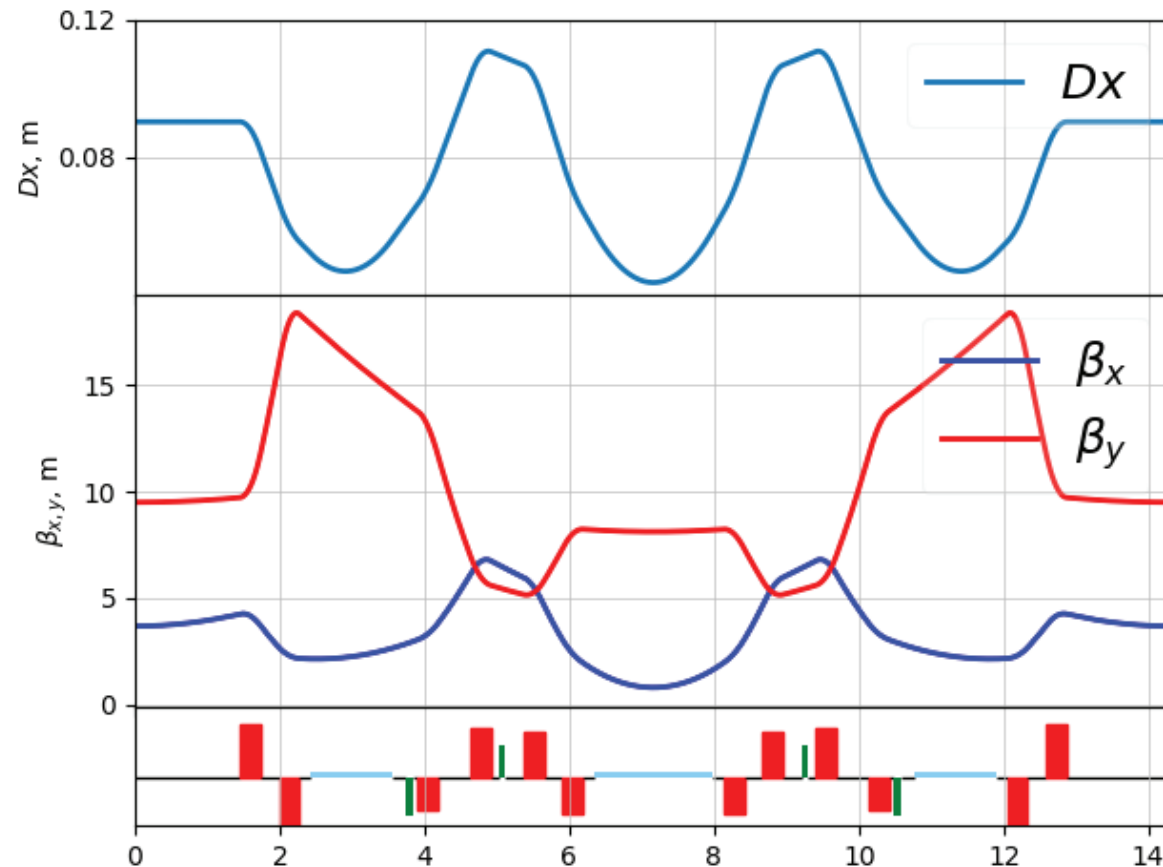


USSR booster synchrotron

Circumference: 515.5 m

Superperiods: 36

Emittance: 4.1 nm·rad



Most likely
on-axis injection
into main storage ring



Design tasks and challenges

- Next rounds of emittance – DA – circumference optimization
- Emittance decrease with the help of Robinson Wigglers
- Round beam mode development
- Beam dynamics simulation taking into account intrabeam scattering, instabilities, field errors and misalignments, influence of IDs
- The innovations and evolution in the technologies of magnetic, vacuum, RF systems, the solution of new problems in materials science and instrument engineering are required
- New ideas in control systems are required
- Work with high precisions



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Thank you and join us!

